Github Actions

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History

Started in 2018

Was originally intended to be support quick shortcuts for github, but evolved to be mainly for continuous integration and continuous deployment (CI/CD) post 2019

What is Github Actions?

On a high level, Github Actions is an automation tool that reacts to certain triggers and runs commands in response.

It is especially helpful for CI/CD. Things like…

- Automating builds
- Testing code
- Deploying code



GitHub Actions

Components of Github Actions

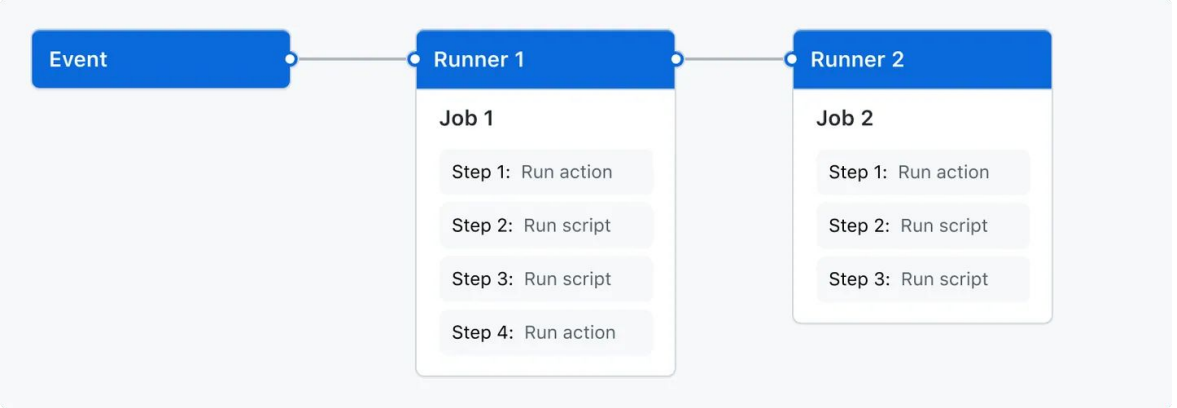
Workflow: Goes into the .github/workflows directory. Defined by a YAML file (e.g. test.yml).

Event: Any sort of trigger in a github repository. Can be creating pull requests, pushing commits, or even be scheduled.

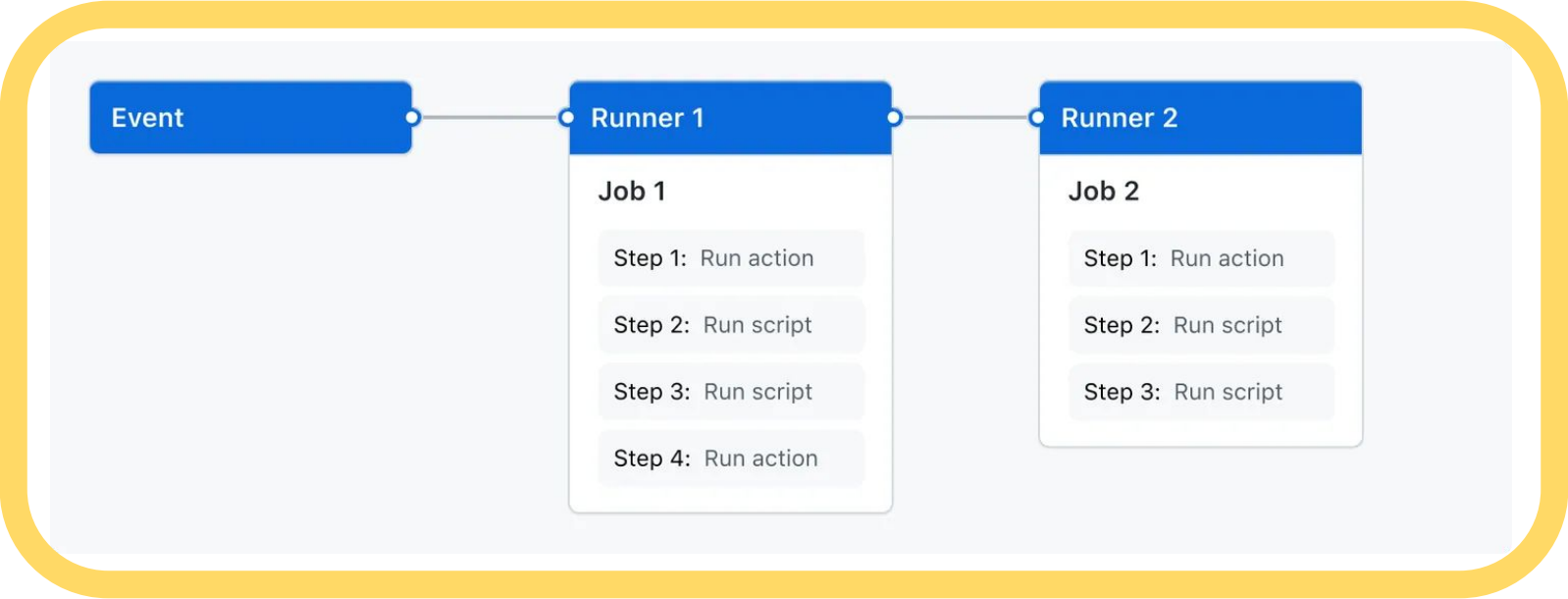
Job: A set of commands/actions to run

- Action: A predefined, reusable set of jobs or code that performs specific tasks (some defaults are provided by github)

Runner: The server that runs the workflows (provided by github)

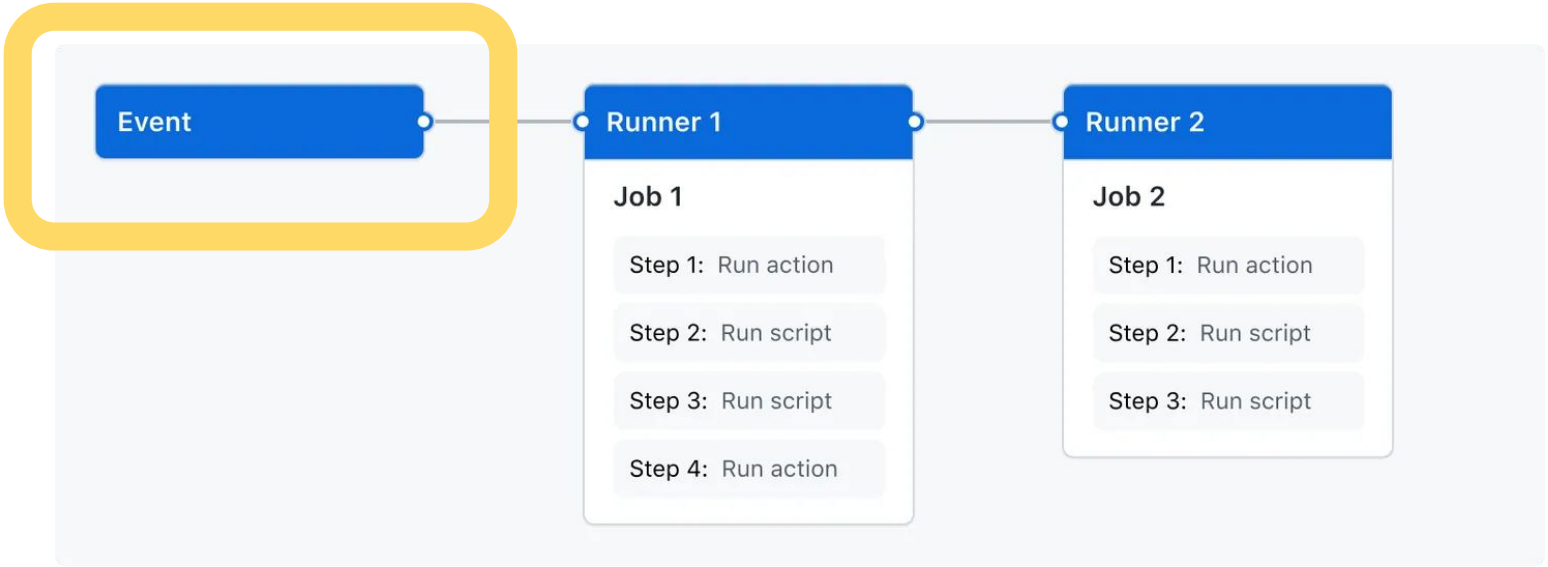


Workflow



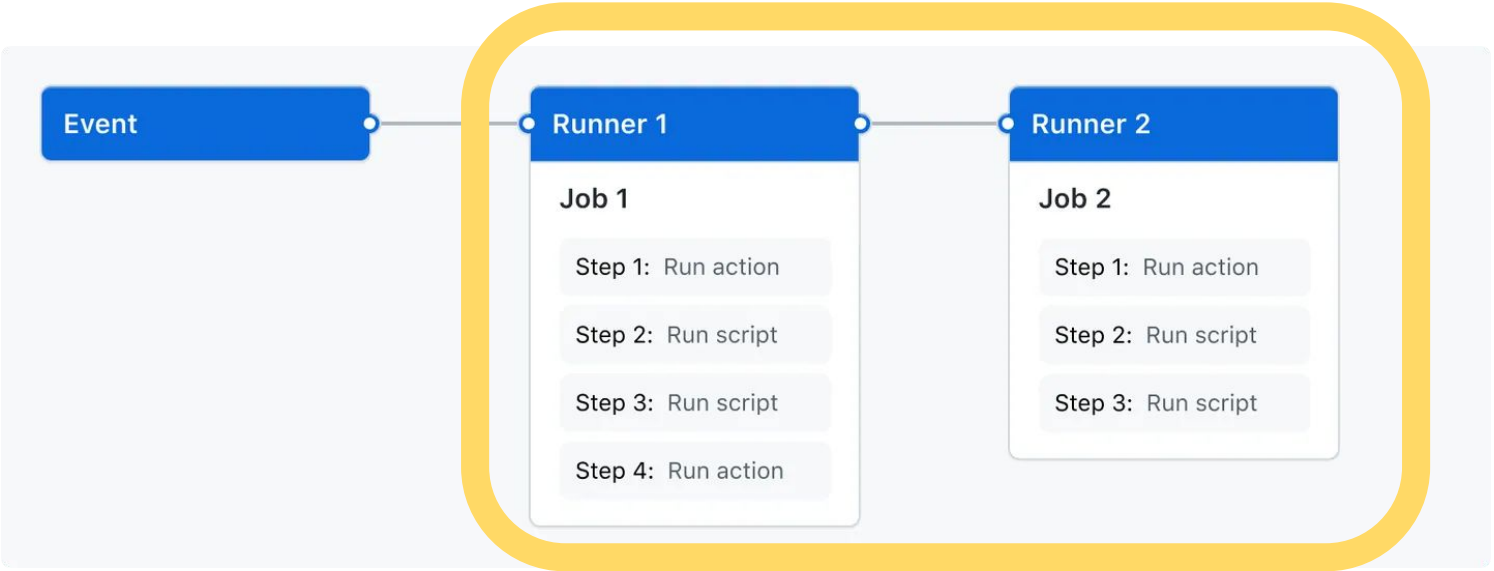
Example: Testing Code

Event



Event = pull request

Runners and Jobs



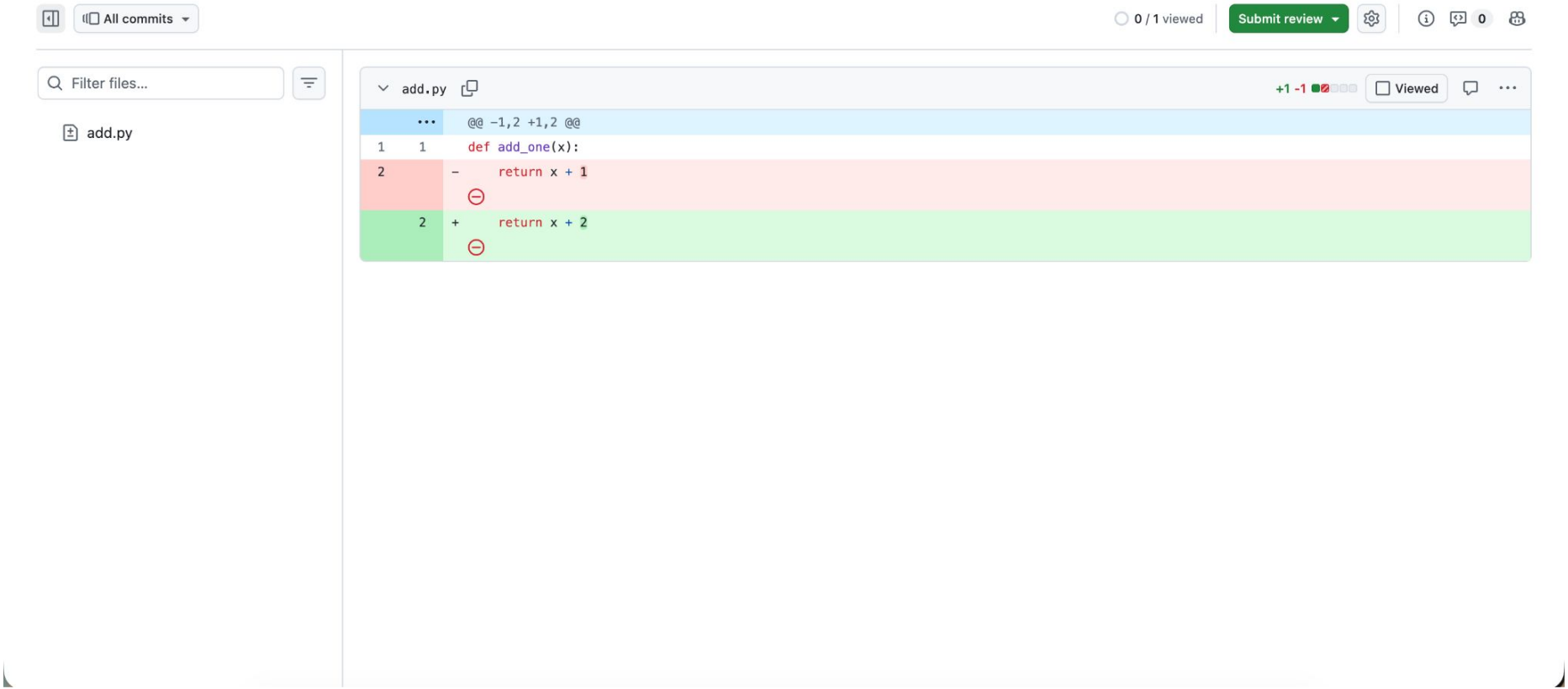
Job 1: Lint
Job 2: Run tests

- Note: Each job gets its own runner. Parallel by default, but can be set to sequential

Demo Time !

https://github.com/0wenSam/Github_Actions_Example

- Private repo


All commits
0 / 1 viewed
Submit review
0
Filter files...
add.py
add.py
+1 -1
Viewed
@@ -1,2 +1,2 @@
def add_one(x):
return x + 1
return x + 2

Files

main

Go to file T

- .github
- add.py
- test_add.py

Github_Actions_Example / test_add.py

0wenSam Commit one — 3658b4b · 30 minutes ago — History

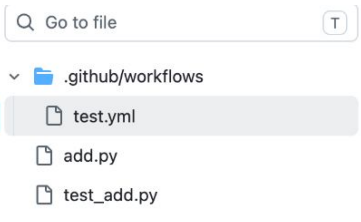
Code | Blame | 11 lines (8 loc) · 234 Bytes | Raw

```python
import unittest
from add import add_one

class TestCalculator(unittest.TestCase):

    def test_add(self):
        self.assertEqual(add_one(0), 1)
        self.assertEqual(add_one(1), 2)

if __name__ == "__main__":
    unittest.main()
```



Code | Blame | 27 lines (21 loc) · 518 Bytes | Raw

```yaml
name: Run Tests

on:
  pull_request:
    types: [opened, synchronize, reopened]

jobs:
  test:
    runs-on: ubuntu-latest

    steps:
      - name: Check out code
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.12"

      - name: Run tests
        run: python -m unittest test_add.py -v
  #If I wanted to have more jobs/runners, I could do something like...
  #lint:
    #runs-on: ubuntu-latest

    #steps:
      #...
```

I made a mistake #1

Merge status | Code

Open | 0wenSam wants to merge 1 commit into `main` from `example`

Conversation 0 | Commits 1 | Checks 1 | Files changed 1 | +1 -1



0wenSam commented 26 minutes ago

No description provided.

I made a mistake — 459bdad

All checks have failed
1 failing check

Run Tests / test (pull_request) Failing after 7s

No conflicts with base branch
Merging can be performed automatically.

Merge pull request | You can also merge this with the command line. View command line instructions.

Still in progress? Convert to draft

Reviewers
No reviews
Still in progress? Convert to draft

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

Summary

- Github actions is used to automate workflows in github.
- Useful for building, testing, and deploying code

Potential use cases in our projects

- My group uses it for continuous deployment of our webpage
- Can also be useful if we ever decide to implement test cases :p

References

https://docs.github.com/en/actions/get-started/understand-github-actions

https://docs.github.com/en/actions/tutorials/build-and-test-code/python

https://github.com/resources/whitepapers/actions

https://techcrunch.com/2019/08/08/github-actions-is-now-a-ci-cd-service/